Evidence map – where every slide comes from

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How to use this document

Every number and claim on the course slides can be traced to one notebook section, one output file and one page of the PDF report. The tables below go slide by slide in presentation order.

- **Notebook section** = the markdown heading inside the notebook, e.g. *Notebook 03 §2* is the cell headed "2. Capital lockup over time".
- **Output file** = the text summary or figure the notebook writes. Summaries are in `outputs/tables/`, figures in `outputs/figures/`. The summary files are plain text: search them for the section title in quotes, e.g. "=== Pareto".
- **Report** = section and page of `report/report.pdf` (11 pages).

Everything is in the GitHub repository Niko-Geo/inventory-management-hslu. The raw data are not in the repository; the summary files and figures are, so every number can be checked without running anything. All money values are anonymised units, so read them as relative figures.

The pipeline: five notebooks, run in order

Each notebook reads the previous one's output, so the chain from raw data to recommendation is linear. Settings (reference date, analysis window, outlier rule, service levels) live in `src/config.py`.

Notebook	Question it answers	Reads	Writes	Report
01 Load and overview	What is in the raw data?	Raw CSV and XLS in `data/raw/`	`transactions.parquet`, `01_overview.txt`	§2, p. 2
02 Clean and demand	Reference date, master data, outliers, monthly demand	`transactions.parquet`	`transactions_clean`, `articles`, `demand_monthly` (Parquet), `02_cleaning_summary.txt`	§3, p. 2–3
03 Data analysis	Where is the capital, and why?	Outputs of 02	`article_features.parquet` (48 metrics per article), `03_analysis_summary.txt`, 7 figures `03_*.png`	§4, p. 3–4

Notebook	Question it answers	Reads	Writes	Report
04 Classification	Which articles belong in stock?	`article_features.parquet`	`article_clusters.parquet`, `04_classification_summary.txt`, 4 figures `04_*.png`	§5, p. 4–5
05 Recommendations	How much stock, tested how, and with what effect?	Outputs of 02–04	`05_recommendations.xlsx`, `05_ten_articles.csv`, `05_recommendations_all_articles.csv`, `05_recommendation_summary.txt`, figures `05_*.png`	§6–7, p. 6–9

The report (`report/report.qmd`, rendered to `report.pdf`) only pulls figures from `outputs/figures/` and numbers from the summary files; it computes nothing itself.

Part 1 – slides 1–5 (colleague): situation, data, where the capital is

Slide 1 (title) carries no claims beyond the headline, which comes from slide 13.

Slide	Claim on the slide	Value	Computed in	Output	Report
2 Situation	Capital lockup rose 2018 → 2021	+41% (3.51 M → 4.94 M)	Notebook 03 §2 "Capital lockup over time"	`03_analysis_summary.txt` → "Capital lockup per year", column lockup_vs_2018_%	§4.1, p. 3, Figure 1
2 Situation	Coverage in days	128 → 136 days	Notebook 03 §2 (lockup ÷ consumption value × 360)	same table, column coverage_days	§4.1, p. 3
2 Situation	Bar chart	–	Notebook 03 §2	`03_capital_lockup_trend.png`	Figure 1, p. 3
2 Situation	Goal, definition of capital lockup	–	Case study PDF, slides 3 and 6	–	§2, p. 2
3 Approach	2.69 M transactions, 17 attributes, 58,764 articles	2,694,849 rows	Notebook 01 §2–3	`01_overview.txt` → "Shape", "Missing values & cardinality"	§2, p. 2
3 Approach	Reference date 21.01.2022	only monthly bookings after it	Notebook 02 §A1	`02_cleaning_summary.txt` → "Transactions per month from 2021-07"	§3, p. 2

Slide	Claim on the slide	Value	Computed in	Output	Report
3 Approach	1,751 stock articles (safety stock > 0)	1,751	Notebook 01 §6	`01_overview.txt` → "Article counts"	§2, p. 2
3 Approach	Demand = sales + usage for production, net of returns	entry types 1 and 5	Notebook 01 §5 (signs per type), `src/config.py` DEMAND_TYPES	`01_overview.txt` → "Entry types"	§2, p. 2
3 Approach	Master data are a snapshot	0% of articles change safety stock, price, lead time, MOQ	Notebook 02 §A2	`02_cleaning_summary.txt` → "Attribute variation per article"	§3, p. 2
4 Capital	Top 10% of articles hold 74% of capital (top 1%: 31%)	74.3%	Notebook 03 §3	`03_analysis_summary.txt` → "Pareto"; `03_pareto_capital_lockup.png`	§4.2, p. 3; Appendix B, p. 11
4 Capital	Articles without demand > 1 year or never hold ~5%	2% + 0% … + 3%	Notebook 03 §7	"Time since last demand", column lockup_share_%	§4.4, p. 4
4 Capital	Semi-finished 49%, retail 36%, finished 12%	–	Notebook 03 §4 (derived from entry types 5 and 6)	"Product type (derived)"	§4.2, p. 3
4 Capital	C articles 401 days coverage, A articles 112 days	–	Notebook 03 §5	"ABC classes", column coverage_days	§4.2, p. 3

Slide	Claim on the slide	Value	Computed in	Output	Report
5 Insight	91 articles with reorder point < 0.5× lead-time demand hold 23%	91 / 23%	Notebook 03 §9	"Current reorder point vs. lead-time demand"; `03_rop_ratio.png`	§4.4, p. 4, Figure 3
5 Insight	441 articles more than 10× too high	441	Notebook 03 §9	same table	§4.4, p. 4
5 Insight	281 articles: one MOQ > 12 months of demand	281	Notebook 03 §9	"MOQ in months of demand"	§4.4, p. 4
5 Insight	57% of stocked capital beyond the reorder-point policy	4.10 M actual vs 1.78 M simulated	Notebook 05 §3, cell after the efficiency frontier	`05_recommendation_summary.txt` → "Ordering behaviour vs. reorder-point policy"	§6.3, p. 7; §1, p. 2
5 Insight	Current reorder points reach only 74% fill rate	0.740	Notebook 05 §3	"Backtest 2021 – all stocked articles", row current	§6.2, p. 6

Part 2 – slides 6–12 (Niko): classification, safety stock and reorder point, ten articles

Slide	Claim on the slide	Value	Computed in	Output	Report
6 Method	7 candidate features for 1,498 articles; lockup, reorder point, lead time excluded	1,498	Notebook 04 §1 (markdown explains the exclusions)	`04_classification_summary.txt` → "Population", "Candidate features"	§5.1, p. 4–5

Slide	Claim on the slide	Value	Computed in	Output	Report
6 Method	CV dropped by correlation filter	r = 0.93 with ADI	Notebook 04 §2	"Highly correlated pairs"; `04_feature_correlation.png`	§5.1, p. 5
6 Method	Forward selection adds production share, rejects recency	silhouette 0.306 → 0.327	Notebook 04 §3	"Forward selection path"	§5.1, p. 5
6 Method	k-means, k = 3 best	silhouette 0.346, Davies-Bouldin 1.12, Calinski-Harabasz 713	Notebook 04 §4	"Evaluation metrics", "Chosen k"; `04_cluster_evaluation.png`	§5.2, p. 5, Figure 4
6 Method	Stability	ARI mean 0.978, min 0.947	Notebook 04 §4	"Bootstrap stability"	§5.2, p. 5
7 Classes	Size, capital share, profile of C1, C2, C3	743 / 474 / 281; 53% / 39% / 5%	Notebook 04 §5	"Cluster profiles", "Cluster x product_type"	§5.3, p. 5
7 Classes	ND class	253 articles, 2.9%	Notebook 04 §6	"Rule-based class ND"	§5.3, p. 5
7 Classes	Strategy per class; 210 make-to-order, 71 observe, 182 phase out; 1,007 stocked	87% of capital stocked	Notebook 05 §1	`05_recommendation_summary.txt` → "Strategy by class"	§5.3, p. 5
8 Formulas	Case formula and fallback	SS = σ; ROP = D/360 × L + SS; usage/4	Case study PDF slide 11; Notebook 05 §2 function `sizing`	–	§6.1, p. 6
8 Formulas	Textbook formula, Q, expected stock	SS = z·σ·√(L/30); Q = max(MOQ, 1 month)	Notebook 05 §2 function `sizing`	–	§6.1, p. 6

Slide	Claim on the slide	Value	Computed in	Output	Report
9 Backtest	925 of 1,007 articles simulated	925	Notebook 05 §3 function `simulate`	"Backtest coverage"	§6.2, p. 6
9 Backtest	Out of sample: parameters from 2019–2020	PARAM_END = 2020-12-31	Notebook 05 §3; `src/config.py`	–	§6.2, p. 6
10 Frontier	Current 74.0% / 1.78 M; case 87.9% / 1.99 M	–	Notebook 05 §3	"Backtest 2021 – all stocked articles", "Reference points"	§6.2, p. 6
10 Frontier	Seven settings; chosen A90/B85/C80 = 90.0% / 2.38 M; flat 90% = 90.3% / 2.45 M; A98/B95/C90 = 94.2% / 3.07 M	–	Notebook 05 §3 "Efficiency frontier"	"Efficiency frontier", "Chosen service-level setting"; `05_efficiency_frontier.png`	§6.2, p. 6, Figure 5
11 Ten articles	Selection rules for the ten	–	Notebook 05 §5 (`pick(...)` calls)	"Ten selected articles", row reason	§7, p. 7
11 Ten articles	Current ROP, new SS and ROP, capital change, fill rates	per article	Notebook 05 §5	`05_ten_articles.csv` (columns current_rop, new_ss, new_rop, lockup_change_%, sim_fill_rate_current, sim_fill_rate_final)	§7, p. 7 (table), p. 7–9 (text)
12 Deep dive	1425485: ~19,000 units/month, lead time 20 days, ROP 5,553 → 23,255, fill 64% → 93%, capital 241 k → 47 k (−80%)	–	Notebook 05 §5; Notebook 03 §3 (top 15)	`05_ten_articles.csv` (mean_monthly, lead_time, lockup_12m, new_lockup); `05_article_1425485.png`	§7, p. 7–8, Figure 6

Note for slide 12: the dotted orange line in the figure sits at about 20,600, not 23,255, because the backtest uses 2019–2020 parameters only; the recommendation uses all of 2019–2021.

Part 3 – slides 13–14 (colleague) and backup slides

Slide	Claim on the slide	Value	Computed in	Output	Report
13 Result	Capital lockup −55%, 4.96 M → 2.23 M	4,958,257 → 2,232,575	Notebook 05 §4	`05_recommendation_summary.txt` → "Capital lockup effect (textbook)", row TOTAL	§1, p. 2; §6.3, p. 7
13 Result	Fill rate 74% → 90%	0.740 → 0.900	Notebook 05 §3	"Backtest 2021" (current) and "Efficiency frontier" (A90/B85/C80)	§6.2, p. 6
13 Result	Breakdown by strategy (stocked 4.35 → 2.18 M, −50%; make-to-order, order on demand, phase out → 0)	–	Notebook 05 §4	"Capital lockup effect (textbook)"	§6.3, p. 7
13 Result	About −63% with order size = 1 month	stocked 1.78 M instead of 2.18 M	Notebook 05 §4	"Order-size lever for stocked articles"	§6.3, p. 7
14 Recommendations	Seven recommendations	–	Synthesis of notebooks 03–05	–	§8, p. 9
14 Recommendations	37 stock articles without lead time	37	Notebook 01 §6, Notebook 03 §8	`01_overview.txt` → "Inventory articles – lead_time"; `03_lead_time.png`	§4.4, p. 4
14 Limits and outlook	Anonymised data, snapshot master data, stock-series breaks, simplified backtest, normal distribution	–	Notebooks 01–05	–	§9, p. 9–10
Backup 1 Robustness	51% of day transitions break; jumps +365 M / −368 M; yearly difference −5% to +13%; correlation 0.95	–	Notebook 03 §1 and §2b; Notebook 02 §A4	`03_analysis_summary.txt` → "Daily stock reconstruction", "Capital lockup per year – two stock methods"	§3, p. 3

Slide	Claim on the slide	Value	Computed in	Output	Report
Backup 2 Clusters	Centroids, PCA, agreement with ABC 0.06, XYZ −0.04, Syntetos–Boylan 0.07, product type 0.45	–	Notebook 04 §4–5	`04_classification_summary.txt` → "Agreement with classic schemes"; `04_cluster_centroids.png`, `04_clusters_pca.png`	§5.2, p. 5; Appendix B, p. 11
Backup 3 Details	Outlier rule, product type, ABC, demand patterns, clustering, z-values, backtest settings	–	`src/config.py`; Notebooks 02 §B, 03 §4–6, 04 §4, 05 §3	–	Appendix A, p. 11

Tracing a number yourself, and where the answers to likely questions are

Example: where does "57%" come from?

1. Notebook 05 §3 replays 2021 with the company's current reorder points and a disciplined order size (one month of demand or the MOQ). Average simulated stock value: 1.78 M.
2. Notebook 03 §3 gives the actual capital lockup of the last 12 months; summed over the same 925 simulated articles it is 4.10 M.
3. 1 − 1.78 / 4.10 = 57%. Printed in `05_recommendation_summary.txt` under "Ordering behaviour vs. reorder-point policy"; explained in report §6.3, p. 7.

Likely question	Short answer	Where it is
Why 1,751 stock articles and not ~3,600 as in the case slides?	Our data are an anonymised version; we apply the case definition (safety stock > 0) to it	Notebook 01 §6–7; report §2, p. 2
Is 74% the real availability today?	No: it is today's reorder points with disciplined order sizes; excess stock covered the gaps in reality	Notebook 05 §3; report §6.2–6.3, p. 6–7
Why not cluster on capital lockup or reorder points?	That is the policy being evaluated – circular	Notebook 04 §1 markdown; report §5.1, p. 4

Likely question	Short answer	Where it is
A silhouette of 0.35 is low	Moderate structure is normal for continuous article data; stability ARI 0.98 is the stronger argument	Notebook 04 §4; report §5.2, p. 5
Why the textbook formula if the case formula is cheaper?	Service level becomes an explicit choice; if 88% is enough the case formula is cheaper	Notebook 05 §3; report §6.2, p. 6
Are 55% savings realistic?	Direction is robust (formula and simulation agree), but it arrives as stock is used up and needs a change in ordering behaviour	Notebook 05 §4; report §6.3 and §9, p. 7–10
Do the stock-level breaks invalidate the results?	No: two stock methods give the same trend, correlation 0.95	Notebook 03 §2b; report §3, p. 3; backup slide 1
How were the ten articles chosen?	By rule, one per situation, highest capital first	Notebook 05 §5; report §7, p. 7

The recommendation for every one of the 1,751 stock articles is in `outputs/tables/05_recommendations.xlsx`, sheet "all_articles".